

2 Phenotypic variation exists in many forms.

(a) Some examples of phenotypic variation in plants and animals are described in Table 2.1.

Complete Table 2.1 by stating whether the cause of variation for each described example is likely to be due to:

- genetic factors, V_G
- environmental factors, V_E
- a combination of genetic and environmental factors, $V_G + V_E$.

Table 2.1

description of phenotypic variation	cause of variation
Tomato plants grown in a glasshouse and grown outside vary in the yield of tomatoes they produce. Seventeen genes associated with tomato yield have been identified.	
New strawberry plants from the variety called Sweet Ann are made by asexual reproduction. The new plants grow to different sizes and produce different numbers of fruit.	
The domestic cat has a blood group system with three possible blood types: A, B and AB. The blood types are determined by antigens present on the cell surface membrane of red blood cells.	
Over 50 genes have variants that are associated with excessive weight gain in humans. Other risk factors for excessive weight gain include diet and exercise.	
Resting heart rate in humans varies between different individuals. Some factors that influence resting heart rate include: biological sex, family history of heart disease, number of cigarettes smoked, medication taken.	

[3]

(b) Name a spontaneous, random event occurring in cells that can be a source of phenotypic variation.

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(c) Other than the event named in (b), describe the features of sexual reproduction that contribute to the production of genetically different offspring.

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[Total: 7]